

RESEARCH ARTICLE

WILEY

How to say? Voice analytics of debt collection strategies

Ziyao Huang | Yutao Yang | Chengcheng Liao | Peiyuan Du

Business School, Sichuan University, Chengdu, China

Correspondence

Peiyuan Du, Business School, Sichuan University, Chengdu, 610065, China.
Email: 986821459@qq.com

Funding information

Funds for Sichuan University to Building a World-class University, Grant/Award Number: SKSYL2019-01; National Natural Science Foundation of China, Grant/Award Number: 71925003

Since limit attention has been paid on debt collection calls and collectors' voice metrics, this study explores how collectors' voice metrics influence borrowers' delinquent behavior in debt collection calls. Results of survival analysis show that collectors' voice with higher pitch, higher speed, lower loudness, and lower entropy of energy prolongs borrowers' delinquent time. Mediation analysis indicates that borrowers' voice energy and call duration mediate the effects of collectors' voice metrics on delinquent time. Additional analysis on heterogeneity suggests that short-term and long-term repayment borrowers respond to collectors' voice metrics differently.

1 | INTRODUCTION

Consumers' demand for unsecured, small-amount and short-term finance has boomed in the last two decades (Banks et al., 2015) as they routinely rely on it for purchases large and small (Kagan & Brock, 2020). However, with the increase of consumer finance, the issue of borrowers' poor payment behaviors appears, which might lead to the small loan companies experiencing a difficult operating environment (Cadena & Schoar, 2011). Thus, due to the time value of finance, it is a prominent concern for small loan companies to ensure the timely collection of consumer finance. Small loan companies will adopt some debt collection methods, such as text messaging, e-mail, and calls, to call back the delinquent debt. Previous studies focus on debt collection methods such as text messaging and e-mail with different debt collection strategies. For example, Bursztyn et al. (2019) stated that a text messaging with a strategy of moral appeal will reduce the proportion of overdue customers and decrease default among the customers with the highest ex-ante credit risk. Results of Social and Behavioral Sciences Team (SBST) (Congdon & Shankar, 2015) show that an email reminder can increase the ratio of student loan repayment. Compared with other debt collection ways, debt collection calls tend to be more effective and are thus preferred because it can communicate with the borrowers directly and bilaterally (Tigue et al., 2012). However, limit attention has been paid to debt collection calls and collectors' voice metrics, forming a research gap.

Pitch, intonation, rate of speech, affective tone of voice, and other voice metrics were always called *paralanguage* in the previous linguistics studies (Wharton, 2004). Pragmatics relates to the meaning

of language and paralanguage within a concrete speech situation and is concerned with what human utterances can perform. Speech act is one of the central concepts in pragmatic studies (Tlidi, 2020). The theory is based on the notion that we use language not only to describe something like people or states of affairs but also to perform a lot of actions. Based on speech act theory, this study exploits the impact of voice metrics of collectors on the delinquent time of borrowers and identifies its underlying mechanism. Voice data are from delinquent borrowers who are randomized to receive debt collection calls from the collection service agents in a small loan company (that wishes to be anonymous). The auditory data were transferred to the structured data through Python packages named *pyAudioAnalysis* and *librosa*. We employ accelerated failure time (AFT) model, cox risk model, and ordinary least square (OLS), respectively, to explore the main effect and ensure the robustness. In addition, the machine learning method named classification and regression tree (CART) is used to identify the differences in the perception of voice among different groups of borrowers.

Results of the AFT model show that the voice metrics of collectors will directly influence the delinquent time of borrowers, and result of cox risk model and OLS model prove the robustness of the main effect. Specifically, higher loudness and entropy of energy (i.e., the degree of energy change in a frame of signal, describing the time domain distribution of the signal) shorten the delinquent time, while higher pitch and speed have the adverse impact. Additional CART results show that the influence of the voice metrics of the collector on the borrowers' repayment behavior is heterogeneous. Moreover, the call duration and energy of borrowers mediate the effects of collectors' voice metrics on borrowers' repayment behavior.

The contributions of this study are summarized as follows. First, as far as we know, this is one of the first attempts to analyze the voice metrics of collectors and borrowers in the debt collection calls. Previous studies focused on collection strategies like text messaging and e-mail while ignoring debt collection calls; thus, this study makes up for the blank of debt collection strategy research. Second, this study applies the speech act theory in pragmatics to the field of economics, which not only expands the field of linguistic application but also provides a new direction for economic studies. Third, this study has identified the effects and underlying mechanism of collectors' voice metrics on borrowers' delinquent time, which can be useful for small loan companies understanding the borrowers' repayment behaviors and then establish more accurate risk control models to improve delinquent debt recovery rate and control loan risk. Finally, this study uses machine learning method to further identify the heterogeneity of borrowers' perception of voice metrics, which provides theoretical guidance for practical applications of small loan companies to distribute the debt collectors reasonably according to the delinquent time and formulate accurate debt collection strategies.

2 | LITERATURE REVIEW

In finance, studies on debt collection mainly focus on the strategies using to call back the delinquent debt that is what to say in the text messaging or email debt collection ways. Reminders and warnings are the two most commonly used ways (Karlan et al., 2012; Visaria, 2009). For the remind strategy, Pekonen (2014) cooperated with a Nordic debt collection agency to explore whether the text message reminds effectively in debt collection and indicated that text message reminders have a positive impact on debt repayment. Also, the text message reminders could have a larger impact on persons who recently have not suffered from payment difficulties than those with recent payment difficulties. This is consistent with the research conclusion of Karlan et al. (2012), Cadena and Schoar (2011), and Johnson et al. (2020) that text messaging or email reminders can effectively increase the repayment rate of delinquent loans. As for the warning strategy, social warning and legal warning are the two widely used types. Bursztyn et al. (2019) used an experiment with the credit card customers of a large Islamic bank in Indonesia to explore the role of morality in debt repayment. And they found that a text messaging with a moral appeal will reduce the proportion of delinquent customers and decrease default among the customers with the highest ex-ante credit risk. Brusa et al. (2019) cooperated with a medium-sized consumer finance loan agency which provides exclusive debt collection services for an online P2P platform in China to test the role of a novel reputation-based incentive in the debt collection. They found that late borrowers are sensitive to the generosity of the monetary incentive but equally responsive to a warning of reporting delinquent behavior to mobile contacts and a full late fee waiver within 24 h. Hermes et al. (2005) investigated group lending programs in Eritrea, and they pointed out that peer monitoring by and social ties of the group leader may help to reduce moral hazard behavior of

group members. In terms of legal warnings, Visaria (2009) affirmed the role of legal proceedings in debt collection and found that the tribunals reduced delinquency for the average loan by 28%. In addition, Fellner et al. (2013) compared the effect of threat treatment, moral appeal, and social information in the mailing reminder through a large-scale natural field experiment. The main findings showed that the comparison across the different mailing treatments reveals a significant effect of the threat treatment. A threat treatment stressing a high detection risk and increased the salience of possible legal and financial sanctions will increase repayment by one additional percentage point. Related studies are summarized in Table 1.

Although there are many studies on debt collection strategies, they mainly focus on the way of what to say, and little attention has been paid to the voice metrics (how to say) of debt collection calls. In linguistics, vocal cues such as rate of speech (speed), intensity (perceived loudness), fundamental frequency (or pitch), and duration are known as *paralanguage* because they can provide information that “surrounds” the linguistic or verbal message. Pragmatics comes as one of the best approaches to study the communicative function of language and paralanguage in use because it is about “how language is used in communication” (Leech, 1983). To understand the communicative function of an utterance is to understand the speech act it performs. Indeed, speech act is one of the central concepts in pragmatic studies. In other words, pragmatic studies are concerned with what human utterances can perform. Therefore, speech act theory is always used to account for one of the central phenomena of any general pragmatic theory. Speech act theory was introduced and developed by Austin (1962) and was later revised by Searle (1969), Ross (1970), Sadock (1974), and others. This theory shows that we use language not only to describe something like people or states of affairs but also to perform a lot of actions that may change the world. Thus, when collectors say to borrowers in the debt collection calls, his or her speech communicates a certain function like to order, to warn, and to persuade. And the speech act implied by the certain paralinguistic features of voice will trigger borrowers' psychological stimulation and thus influence social behavior (Ketrow, 1990). Therefore, we speculate that the voice metrics of collectors will affect the delinquent time of borrowers.

3 | HYPOTHESES DEVELOPMENT

The important influence of pitch has been proven in speech (Klofstad et al., 2015; Tigue et al., 2012), psychology (Apple et al., 1979) and other fields. Studies show that the speakers with high-pitched voices are often considered to be less truthful, emphatic, and potent and more nervous. In addition, lower-pitched voices make speakers feel stronger, having greater integrity and physical prowess. Therefore, we speculate that collectors with lower-pitched voice may be more trustworthy and thus promote repayment. On the basis, we propose the following hypothesis:

Hypothesis 1a. *A collector with lower-pitched voice will urge borrowers shorten the delinquent time.*

TABLE 1 The literature review of debt collection strategies

Author (year)	Communication tools				Strategy		Data type		Dependent variable		
	SMS	Email	Lawsuit	Phone call	What to say	How to say	Unstructured	Structured	Repay or not	Repayment rate	Repayment time
Karlan et al. (2012)	✓				✓			✓			✓
Bursztyn et al. (2019)	✓				✓			✓	✓		
Cadena and Schoar (2011)	✓				✓			✓	✓		
Johnson et al. (2020)	✓				✓			✓	✓		
Social and Behavioral Sciences Team (2015)		✓			✓			✓	✓		
Pekonen (2014)	✓				✓			✓		✓	
Moulton et al. (2015)	✓				✓			✓		✓	
Brusa et al. (2019)	✓				✓			✓	✓		
Eijkel et al. (2011)	/	/	/	/	✓		/	/	✓		
Visaria (2009)			✓		✓			✓		✓	
Fellner et al. (2013)		✓			✓			✓	✓		
This study			✓			✓	✓				✓

According to Ketrow (1990), loudness has an impact on persons' perception. In addition, Gélinas-Chebat et al. (1999) found that high intensity has better effects on the message quality perception than low or moderate intensity. The study of Chebat et al. (2007) confirmed this point again that a moderate intensity is associated with a more credible source (Chebat et al., 2007). Based on previous studies, we speculate that the collectors with louder voice will transmit information more accurate, thus promoting the borrower's delinquent time. Therefore, we propose the following hypothesis:

Hypothesis 1b. *A collector with louder voice will urge borrowers shorten the delinquent time.*

Gélinas-Chebat et al. (1999) pointed that fast speech rate (speaking speed) enhances message execution and credibility, because the message is perceived as being well done and more credible when the speaker speaks fast. Besides, Guyer et al. (2019) stated that the fast speech rate is always associated with confidence speakers. Taken together, we thus hypothesize that the speed of collectors will influence the delinquent time.

Hypothesis 1c. *A collector with a higher speaking speed will urge borrowers shorten the delinquent time.*

Ketrow (1990) found that a fluent voice using a standard dialect, clear diction, and a "normal" voice quality lends itself to greatest perceptions of credibility and social attractiveness for the speaker. This in turn should lead to greater persuasiveness and compliance on the part of the target. Moreover, voice continuity could enhance the ability of listeners' auditory attention and the effect is more obvious when the listener hears two different numbers (Bressler et al., 2014). Therefore, the borrowers' auditory attention is more likely to be attracted by the collectors with high speech continuity (i.e., entropy of energy). We propose the following hypothesis:

Hypothesis 1d. *A collector with a stronger speech continuity will urge borrowers shorten the delinquent time.*

In addition, since debt collection call is a two-way communication between collectors and borrowers (Yu et al., 2005), responses of borrowers should also be considered in the debt collection calls. Energy is the sum of squares of each frame signal which reflects the strength of the signal energy. In general, the more continuous the speech, the greater the energy (Bachu et al., 2008). Borrowers with high level of energy are more active to communicate in the debt collection call. In fact, many studies in marketing have yielded a significant relationship between the communication initiative and customer purchases (Coppett & Voorhees, 1985). Thus, borrowers who actively respond to the collectors in the debt collection call may have shorter repayment time. Moreover, the call duration reflects the communication willingness of the borrower (Gil et al., 2014). The longer call duration indicates that the borrower and the collector repeatedly confirm the

loan information, which may affect the subsequent repayment behavior. We therefore posit that:

Hypothesis 2a. *The voice energy of borrowers mediates the effects of the voice metrics of collectors on the delinquent time of borrowers.*

Hypothesis 2b. *The call duration of borrowers mediates the effects of the voice metrics of collectors on the delinquent time of borrowers.*

The model specification is shown in Figure 1.

4 | METHODOLOGY

4.1 | Data

The dataset is obtained from a small loan company (that wishes to be anonymous) which is in China and has served more than tens of millions of people. Consumers can view loan information and place orders on the app,¹ which is developed by this company. Generally, these loans are used to buy electronic products such as mobile phones or computers. The company employs hundreds of debt collectors to handle debt collection through debt collection calls. Details of borrowers and the app are introduced in Appendix A.

In order to enhance the robustness of results, we take two measures to control the heterogeneity of the data samples when extracting samples from the data bases. Firstly, the dataset is from the auditory data pool in May and June 2019 because only in these 2 months, the company adopted a single telephone collection strategy without superimposing other collection strategies, which can exclude the impact of other collection strategies on the repayment behavior of borrowers. Secondly, the borrowers' telephone numbers are randomly assigned to the debt collectors who are responsible for the debt collection. It is helpful for us to ensure fairness in practice. Based on the above measures, the whole dataset concludes 10,666 audio recordings of debt collection calls.

Next, all the delinquent borrowers receive the debt collection call on the first overdue day to ensure that there are no differences of the delinquent severeness of borrowers. Finally, there is only one debt

collector responsible for its collection in the first overdue stage of delinquent borrowers so that we could eliminate the interference of other debt calls on the delinquent days. There are 1600 data samples left after taking these two measures, which account for 15% of the total data. After eliminating borrowers with duplicate recordings and missing demographic features, there are a total of 1562 complete data sample be used. The data generation process is shown in Figure 2.

4.2 | Variables

Dependent variable: *delinquent time*. Previous studies mainly focus on the repayment rate (Pekonen, 2014; Moulton et al. 2015; Visaria, 2009). However, the repayment rate of small loan companies is often stable because of the application and character of micro consumer finance. For example, according to NBC News (2018), the average repayment rate for small loans offered by the industry is higher than 95%. In China, the national small loan repayment rate is also as high as 91% in the first half of 2020 (The People's Bank of China, 2020). Compared with the repayment rate, delinquent time may be a more important focus. It is significant for small loan companies to handle the overdue time of delinquent borrowers because sufficient funds can help small loan companies to experience a healthy operating environment (Cadena & Schoar, 2011). Therefore, we use the delinquent time of borrowers as the dependent variable, which is defined as the number of days between the debt collection calls date and the repayment date.

Independent variables: voice metrics of collectors. Collectors' voice metrics including *pitch*, *loudness*, *speed*, and *entropy of energy* are used as independent variables. Pitch, loudness, and speed are the three common factors in voice analysis (Apple et al., 1979), and some studies have shown that the intensity, intonation, and speech rate of telesales can influence the customers' behavior; therefore, we take these three factors into our consideration. Besides, the entropy of energy means the entropy of subframes' normalized energies (Papakpistas et al., 2017). It is usually used to judge the continuity of speech (Bachu et al., 2008). Generally, the higher the energy entropy, the longer the silence time and the weaker the speech continuity (El Ayadi et al., 2011). The extraction process of voice metrics is shown in Appendix B.

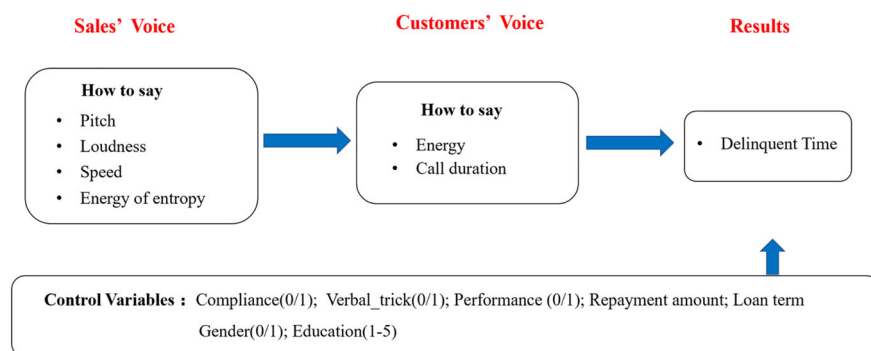
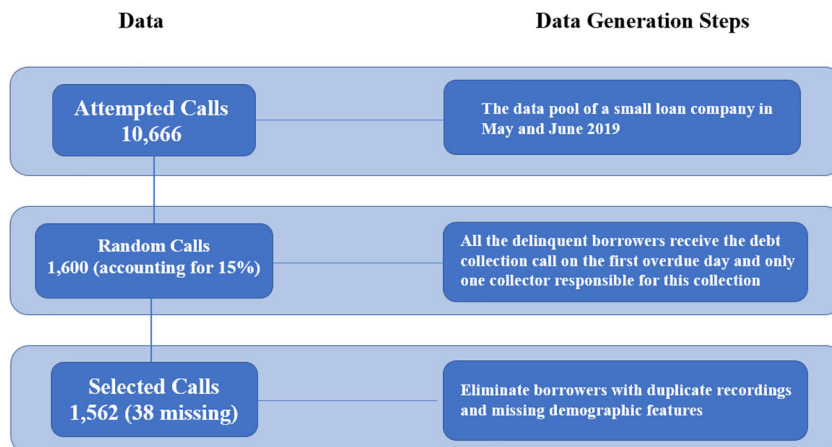


FIGURE 1 Conceptual framework [Colour figure can be viewed at wileyonlinelibrary.com]

FIGURE 2 Data generation process [Colour figure can be viewed at wileyonlinelibrary.com]



Mediators: voice metrics of borrowers. Because that the speech continuity and duration often represent the engagement of borrowers (Gil et al., 2014), which will affect the subsequent repayment behavior, we extract the energy and call duration of borrowers. The energy is the indicator to describe the speech signal, which can help to judge the continuity of speech (Bachu et al., 2008). The call duration is the voice call time of the borrowers, measured in seconds.

Control variables: control variables include debt collection strategy, borrowers' response, loan information, and demographic factors. Those control variables are important in an organizational researcher's methodological toolkit (Bernerth & Aguinis, 2016). Further, those factors should logically be considered as parts of the explanation for the effects of collectors' voice metrics on the delinquent time of borrowers. In this situation, we mainly discuss how the collectors say in the call, that is, the impact of voice metrics on the delinquent time that should eliminate the interference of what the collectors say in the debt collection call. Therefore, to ensure that the model estimations are unbiased and reliable, four kinds of variables are used as control variables. (1) *Verbal trick*: Collectors usually put pressure on severe overdue borrowers to prompt them to repay early. For borrowers who are only overdue for a few days, collectors just remind them using courtesy as usual. Therefore, we use the verbal trick (pressure vs. polite) as a control variable, with the pressure trick coded as 1 and the polite trick coded as 0. (2) *Compliance*: When receiving the collection call, borrowers usually response in two ways, one is to show obedience and promise to repay as soon as possible; the other is to show denial and find excuses to explain. We use a variable named *compliance* to indicate this phenomenon, that is, when the borrowers say positive words such as "OK, I will", it is coded as 1, and when it says negative words such as "No, I won't," it is coded as 0. (3) *The loan information*: Loan information includes repayment amount and loan term. *Repayment amount* is the amount of the loan repaid by the borrowers, and *loan term* is the loan periods of the borrowers. (4) *Demographic factors*: To ensure the integrity of data to the greatest extent, we choose borrowers' demographic factors including *gender* and *education* as the control variables. All variables used and the descriptive statistics are shown in Table 2.

4.3 | Model specification

4.3.1 | AFT model

The period from loan to repayment of the borrowers can be regarded as the cycle of a person's life path; thus, survival analysis was used. The AFT model is appropriate for predicting the relationship between the collectors' voice metrics and the borrowers' delinquent time. In survival analysis, most researchers use cox risk model to simulate the relationship (Hunjun, 2003). However, cox risk model requires that the effect of covariates must be transformed from risk to survival time. When the survival time is exponential distribution, the transformation becomes very complex (Tian et al., 2014). Thus, some studies used the AFT model for this situation, which is simple, intuitive, and easy to understand. AFT model expresses the failure time directly, while the cox proportional risk model expresses the probability. Therefore, the AFT model is a good alternative to the proportional risk model (Jin et al., 2003; Ma & Bechinski, 2009). In addition, because the AFT model describes the linear relationship between the logarithm of failure time and covariates, it simplifies the steps of the model and reduces the error amplification caused by connecting different models (Cox & Oakes, 1984; Kalbfleisch & Prentice, 2002). The AFT model constructed in this study is as follows:

$$\log t_i = \beta * X + \delta * \varepsilon_i \quad (1)$$

where t_i represents the delinquent period of the borrowers and X is the covariate vector including the *pitch*, *speed*, *loudness*, and *entropy of energy* of the collectors. In addition, we take the exogenous variables' impact into consideration and include the control variables (*verbal trick*, *compliance*, *repayment amount*, *loan term*, *gender*, and *education*) into the AFT model. σ is a scale parameter, and ε_i is a random error.

5 | EMPIRICAL RESULTS

This section discusses the empirical results of our analysis. The first part of the analysis focuses on the main effect that is the impact of

TABLE 2 Variables and descriptive statistics

Variables	Data type	Definition	Min	Max	Mean	SD
Panel A: independent variable						
Pitch	Continuous	How high a collector's voice is, indicating the subjective characteristic of a sound that determines its position in a musical scale	1347	2721	2125	203
Loudness	Continuous	How louder a collector's voice is, describing the magnitude of the auditory sensation produced by physical disturbances, measured by decibel	47	60	56	1.4
Speed	Continuous	How fast a collector's voice is, indicating the speed rate of the collector, measured by the number of words per second	0.4	6	3	0.9
Entropy of energy	Continuous	How strong a collector's speech continuity is, which is the degree of energy change in a frame of signal, describing the time domain distribution of the signal	2	3	3	0.04
Panel B: mediator						
Energy	Continuous	How strong a borrower's speech continuity is, which is the short-term energy, reflecting the strength of the signal	0.002	0.4	0.07	0.05
Call duration	Continuous	Borrowers' call length, recorded by second	0.000	525	28	33
Panel C: control variables						
Verbal trick	Binary	The collection strategy of the collectors, a dichotomous variable with 0 represents polite remarks and 1 represents pressure remarks	0	1	0.3	0.5
Compliance	Binary	The response of the borrowers, a dichotomous variable with 0 represents that borrower show compliance and 1 represents not.	0	1	0.04	0.2
Repayment amount	Continuous	The actual repayment amounts of borrowers	85	4,276	1,087	744
Loan term	Integer	The loan periods of the borrowers	3	31	16	5
Gender	Binary	The gender of borrowers	0	1	0.2	0.4
Education	Category	The education degree of borrowers	0	5	3	0.4
Panel D: dependent variable						
Delinquent time	Integer	The period from the bill due date to the actual repayment date	10	4	90	14

voice metrics of collectors on the delinquent time of borrowers. The second part is the machine learning results of the heterogeneity effect. The third part shows the mediating effects of the voice metrics of borrowers.

5.1 | Randomization check

Based on random collection case allocation and random data extraction, we randomly control unobserved variables such as heterogeneity of borrowers. We use the median to divide the overdue days into two groups and test whether there are significant differences in the voice characteristics of different groups. Table 3 shows the results of the independent-sample *T* test. There was no significant difference between the two groups of *pitch* ($T = 11.706$, $p = 0.495$), *loudness* ($T = 13.566$, $p = 0.942$), *speed* ($T = 11.212$, $p = 0.2$), *entropy of energy* ($T = 12.807$, $p = 0.624$), and control variables, indicating that the data were in accordance with the characteristics of randomization. The specific randomization results are shown in Table 3.

5.2 | Main effect

The results of survival analysis and the robustness check are showed in Table 4. It shows that the coefficient for *pitch* ($\beta = 0.477$, $p < 0.1$) is positive and significant at the 10% level, indicating that collectors with higher level of pitch will make the delinquent time longer. Hypothesis 1a is verified. Besides, there is also one indicator that affects the delinquent time of borrowers in the positive direction, that is, the *speed* ($\beta = 0.0312$, $p < 0.01$). This shows that the faster the collector speaks, the longer the borrowers will be deliquated. Hypothesis 1c is verified. In addition, the *loudness* ($\beta = -0.0285$, $p < 0.01$) and *entropy of energy* ($\beta = -1.76$, $p < 0.01$) are negatively influencing the borrower's delinquent time. It means that the collector that has a louder and discontinuous voice will urge the borrowers repay as soon as possible. Hypothesis 1b and Hypothesis 1d are thus supported. Moreover, the *verbal trick*, the *repayment amount*, and the *loan terms* of borrowers also influence the delinquent time. Specifically, for borrowers who are more willing to say "no," their delinquent time is indeed longer. And the more they repay, the shorter their delinquent time.

TABLE 3 Randomization check

Group	N	Pitch	Loudness	Speed	Entropy of energy	Verbal trick	Compliance	Performance	Repayment amount	Loan term	Gender	Education
Long overdue	1153	2120	55.97	2.94	3.1	/	/	/	1099.02	16.66	/	/
Short overdue	409	2142	55.85	2.98	3.1	/	/	/	1054.82	16.35	/	/
T-value		11.706	13.566	11.212	12.807	4.326	3.254	1.63	0.007	4.175	8.419	0.176
P-value		0.495	0.942	0.2	0.624	0.31	0.369	0.334	0.243	0.17	0.141	0.322

In order to check the robustness of the results, we also used cox risk model and OLS. Results of OLS are consistent with the AFT model. In the AFT model, the coefficient of *pitch* is positive, which indicates that collectors with high pitch lead to the long delinquent time of the borrowers. In the cox risk model, it can be understood as the borrowers with longer survival time, which just shows that the risk coefficient of such borrowers is lower. Therefore, the coefficient of this variable in the cox risk model is negative. The results of the above three models show that the main effect is robust.

5.3 | Machine learning with voice metrics for debt collection calls

In consideration of the differences in the perception of voice between groups of people, we expect that borrowers will response differently to the voice metrics. By applying widely available machine learning methods, that is, decision tree, we identify the heterogeneity of borrowers. Decision tree is an excellent machine learning algorithm. It can be used as classification algorithm or regression model because it is easy to understand and realize. Decision tree is a prediction model in machine learning, which represents a mapping relationship between object attributes and values. The common decision tree algorithms are ID3 decision tree, C4.5 decision tree, and CART. In this section, the CART, including the partition of the dividing dimensions along with estimated treatment effects, was then used to describe the heterogeneity of borrowers because it uses Gini coefficient to calculate classification features and its dichotomy can simplify the scale of decision trees and improve the efficiency of generating decision trees. The dimensions used were identified in our analyses of main effects. The CART model is constructed as follows:

$$\text{Gini}(D) = \sum_{k=1}^K p_k(1-p_k) = 1 - \sum_{k=1}^K p_k^2 \quad (2)$$

where p_k is the proportion of class k in sample D . Cart algorithm is a binary classification algorithm, and the calculation formula of Gini coefficient is as follows:

$$\text{Gini}(D|A) = \frac{|D_1|}{D} * \text{Gini}(D_1) + \frac{|D_2|}{D} * \text{Gini}(D_2) \quad (3)$$

We use the median of the repayment time (7 days) as the dividing dimension to discuss the heterogeneity effect of short-term and long-term repayment. The algorithm starts from the root node and builds cart decision tree by recursive training set. The generation of decision tree is divided into five steps. Firstly, for the data set of the current node as D , if the number of samples is less than the threshold or there is no feature, the decision subtree is returned, and the current node stops recursion. Secondly, calculating the Gini coefficient of the sample set D . If the Gini coefficient is less than the threshold, return to the decision tree subtree, and the current node stops recursion. Thirdly, calculating the Gini coefficient of the eigenvalues of the

	AFT	Cox risk	OLS
Pitch	0.477 (0.032)**	−0.9125 (0.154)	21.54 (0.01386)*
Loudness	−0.0285 (0.000)***	0.06752 (0.000796)***	−1.32 (0.000)***
Speed	0.0312 (0.002)***	−0.05804 (0.04446)*	0.8518 (0.03151)*
Entropy of energy	−1.76 (0.000)***	3.591 (0.0000)***	−89.64 (0.000)***
Compliance	0.0239 (0.608)	−0.1072 (0.426209)	0.6589 (0.72288)
Verbal trick	0.146 (0.000)***	−0.020652 (0.000)***	3.157 (0.00265)**
Performance	0.175 (0.000)***	−0.285 (0.000)***	5.082 (0.000)***
Repayment amount	−0.0000591 (0.000)***	0.000956 (0.003763)**	−0.002703 (0.000)***
Loan term	−0.00308 (0.078)*	0.005903 (0.246209)	−0.08006 (0.24960)
Gender	−0.0183 (0.407)	0.07281 (0.251384)	0.2149 (0.80541)
Education	0.0253 (0.407)	−0.08139 (0.354520)	−0.271 (0.82324)
Debt collectors	Y	Y	Y
P (model)	4.6e-26	3e-10	2.2e-16
F-test	−1571.1		

Abbreviations: AFT, accelerated failure time; OLS, ordinary least square.

* $p < .1$. ** $p < .05$. *** $p < .01$.

TABLE 4 The results of survival analysis

existing features of the current node to the dataset D . The expression of Gini coefficient is as follows:

$$Gini(D) = 1 - \sum_{k=1}^K \left(\frac{|C_k|}{|D|} \right)^2 \quad (4)$$

The expression of Gini coefficient here is that for a given sample D , suppose there are k categories, and the number of k categories is C_k .

Fourthly, among the calculated Gini coefficients of each feature value of each feature to the data set D , the feature A with the smallest Gini coefficient and the corresponding feature value a are selected. According to this optimal feature and optimal feature value, the data set is divided into two parts $D1$ and $D2$, and the left and right nodes of the current node are established at the same time. The data set D of the node is $D1$, and the data set D of the right node is $D2$. Finally, calling the left and right child nodes recursively with the four steps mentioned above to generate the decision tree.

The results of decision trees are showed in Figure 3. When the entropy of energy is lower than 3.023, the overall repayment rate within 7 days reaches 35.6%. On the contrary, the repayment rate within 7 days is as high as 75% when the entropy of energy is greater than 3.023. It means that collectors should strengthen the continuity of speech. In addition, if the entropy of energy is less than 3.023 for people whose repayment amount is less than 416.5, the repayment rate within 7 days will drop to 0. When the energy entropy is less than 3.023 and the repayment amount is greater than 416.5, the pitch and speed should be controlled. When the pitch is greater than 1932.393 Hz and the speed is lower than 2.826, the repayment rate within 7 days is the highest, reaching 84.6%.

When the energy entropy is greater than 3.023, the overall repayment rate is 75%; however, the repayment rate is only 12.5% for borrowers whose repayment amount is less than 67.57. Moreover,

loudness is an important influence impact when the repayment amount is between 67.5 and 3372. When the loudness is greater than 56.579 dB, there is a significant collection effect because the repayment rate is 80.7%.

5.4 | Mediating effect

Bootstrap method in the structural equation model was used to test the mediating effect, since the bootstrap is better to test mediating effects than multi-step mediating tests (Taylor et al., 2008). After controlling variables including *verbal trick*, *compliance*, *repayment amount*, *loan term*, *gender*, and *education*, results of mediation analysis show that:

1. *Pitch* influences the *call duration* positively ($\beta = 64.249$, $p < 0.01$), which means that higher pitch of collectors leads longer call duration of borrowers. Also, call duration of borrowers ($\beta = 0.067$, $p < 0.01$) positively affects their delinquent time, which means that longer call duration extends delinquent time. On the contrary, *pitch* influences borrowers' *energy* in a negative direction ($\beta = -0.41$, $p = 0.276$), indicating that the higher the *pitch* of collectors, the weaker the speech continuity of the borrowers. However, this effect is not significant. Furthermore, the *energy* ($\beta = -8.242$, $p < 0.1$) of borrowers negatively affects their delinquent time; that is, the stronger speech continuity of the borrowers leads to shorter delinquent time.
2. *Loudness* influences the *call duration* ($\beta = -19.496$, $p < 0.05$) and *energy* negatively ($\beta = -0.012$, $p = 0.457$), representing that collectors with louder voice make the borrowers speak less and discontinuous. However, the negative effect of the latter is not significant. Call duration ($\beta = 0.067$, $p < 0.01$) and energy

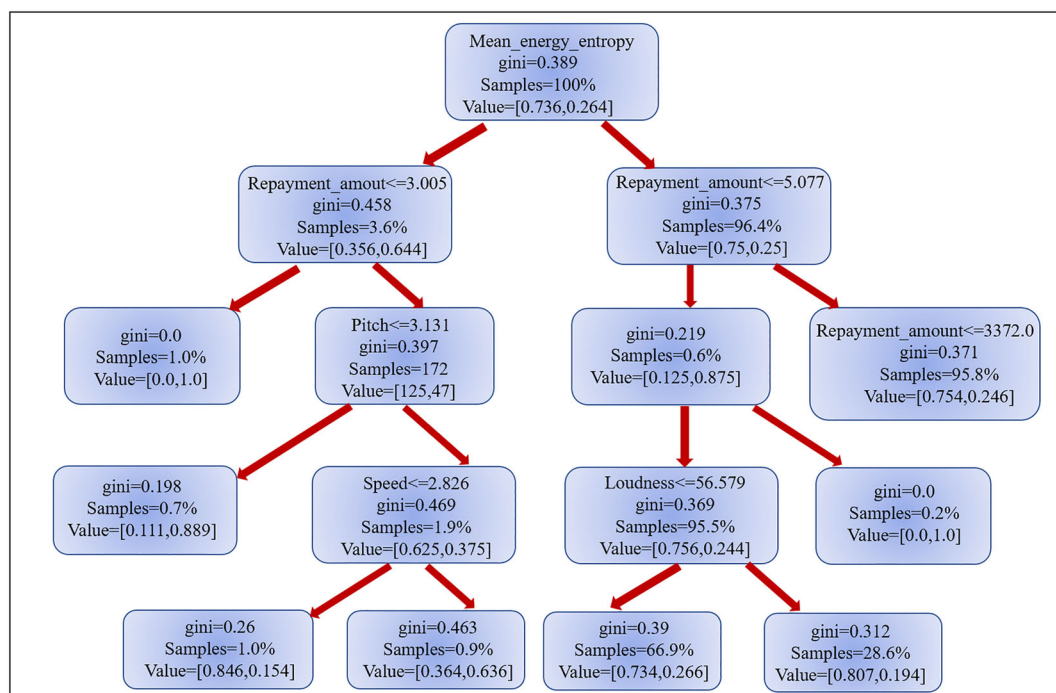


FIGURE 3 The decision tree result of borrowers [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

- ($\beta = -8.242$, $p < 0.1$) of borrowers influence the delinquent time in a positive and negative directions, respectively.
3. Speaking *speed* of the collectors influences the *call duration* ($\beta = -64.344$, $p < 0.01$) and *energy* ($\beta = -0.092$, $p < 0.01$) of borrowers in a negative direction. It shows that a collector with faster speaking speed makes a weaker speech continuity and shorter call duration of borrowers.
 4. *Entropy of energy* negatively affects the *call duration* of borrowers ($\beta = -62.663$, $p < 0.01$), which means that the more the collector says in the voice call, the less the borrower talks and the shorter the call duration. In addition, there is a positive relationship between the collectors' *entropy of energy* and the borrowers' *energy* ($\beta = 0.106$, $p < 0.05$), indicating that in the collection call, the borrower speaks more than the collector. Results of mediating effect are shown in Figure 4.

6 | DISCUSSION AND IMPLICATIONS

6.1 | Conclusion

The rapid development of unsecured, small-amount, and short-term finance industry brings a boom of small loan companies. However, poor payment behavior of borrowers may cause small and even larger businesses in a difficult operating environment (Cadena & Schoar, 2011). The timely collection of delinquent borrowers thus becomes an important issue for small loan companies. Previous studies mainly focused on debt collection methods such as text messaging or e-mail with collection

strategies such as reminders and warnings (Karlan et al., 2012; Bursztyn et al., 2019; Cadena & Schoar, 2011; Johnson et al., 2020; Pekonen, 2014), while ignoring the most commonly used way of debt collection, the debt collection calls. Further, limit attention has been paid to the voice metrics under the situation of the debt collection calls. Therefore, based on the speech act theory, this research explores the influence of the voice metrics of the collectors on the delinquent time of the borrowers and discuss its underlying mechanism.

First, results show that the voice metrics of the collectors affect the delinquent time of the borrowers. Specifically, loudness and entropy of energy of collectors influence the delinquent time in a negative direction. A speaker with louder voice will be seen oppressive (Black, 1961); as a result, a continuous speech with a louder voice will urge borrowers repay as soon as possible. In addition, pitch and speed will prolong the delinquent time. It has been shown that a fast speech rate may cause the listener to ignore the focus (Koomen, 1995). Moreover, the higher pitch seems to result in a speaker being judged as less benevolent (Brown et al., 1974). Therefore, the combination of high pitch and fast speed can easily lead to the anxiety of borrowers (Key & Carr Payne, 1981), reduce the collection effect, and then extend the delinquent time.

Second, results of CART decision tree show that there is a heterogeneity effect in the process of voice metrics influence the delinquent time. The repayment rate with high entropy of energy is significantly higher than that with low entropy of energy. Increasing the loudness of the voice and slowing down the speech speed can prompt borrowers to pay back as soon as possible within 7 days when the entropy of energy is in the low level. This is because when the entropy of energy stays in the low level, the collectors may have

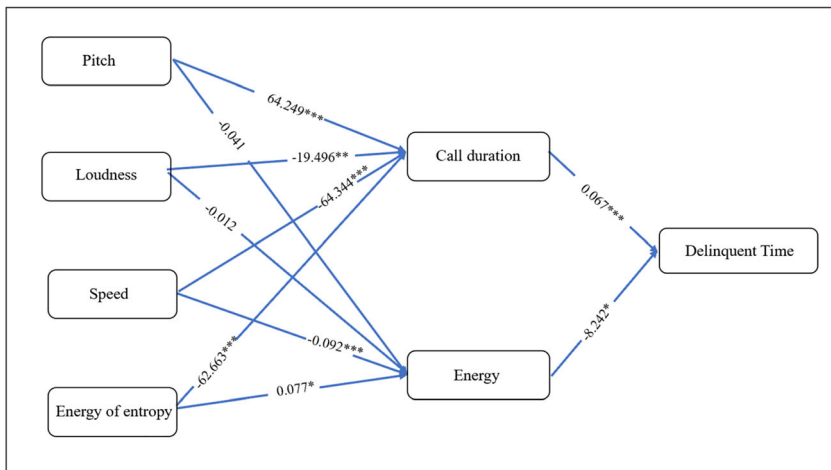


FIGURE 4 Mediation model estimation
[Colour figure can be viewed at
wileyonlinelibrary.com]

multiple rounds of dialog with the borrowers, resulting in a discontinuous state of speech. At this time, the borrowers' reception of louder voice has better effects on the message quality perception than low or moderate loudness (Gélinas-Chebat et al., 1999). In addition, slow speech rate enhances message execution and credibility because the message is perceived as being well done and more credible when the speaker speaks slow. On the other hand, loudness is an important factor when the entropy of energy is in a high level. There will be an excellent collection effect when the collectors speak louder.

Third, the effects of collectors' voice metrics on the borrowers' delinquent time are mediating by the borrowers' voice metrics (call duration and energy). Among them, pitch influences the call duration and energy in the positive direction. In general, the collector usually adopts a pressure strategy when he or she has a higher pitch. In this condition, the call duration will be longer because the borrowers usually doubt or express dissatisfaction. In addition, loudness, speed, and entropy of energy will negatively affect the call duration and energy. Because the people who speak fast will be more fluent, more persuasive, and more active (Apple et al., 1979), a loud voice coupled with people who speak fast and fluently will make the borrowers more reluctant to express and speak less. In the path of mediating variables influencing dependent variables, the call duration of borrowers will prolong the delinquent term. And the energy will make the delinquent time shorter. This may be related to the particularity of the debt collection calls. By listening to the collection calls, we find that borrowers with long call duration are arguing that they do not have the ability to repay or in doubt about the time and amount of repayment mentioned by the collector. This kind of borrowers usually defaults on the delinquent time and need to call for collection again. This is consistent with the conclusion of Du et al. (2020); they stated that emphasizing the adverse consequences of failure to repay loans does not have enduring effects.

6.2 | Theoretical implications

First, by extracting the voice metrics of the collectors in debt collection calls, this study explores the impact of collectors' voice metrics

on the delinquent time of borrowers and thus makes up for the research gap of the existing studies on the debt collection calls. Also, this study combines linguistics and economics and introduces the speech act theory in pragmatics into economics and thus extends both the linguistic and economic studies.

Second, this study reveals the heterogeneity of the impact of voice metrics on repayment behavior through the machine learning method, which introduces a new method into the voice analysis in economics. It provides a new perspective for the combination of AI research methods and economic issues. Moreover, this study provides a theoretical basis for small loan companies to formulate targeted and accurate debt collection strategies.

Third, this study reveals the mediating effects of the borrowers' voice metrics and explains the underlying mechanism of the effects of collectors' voice metrics and thus deepens the speech act theory and provides evidence for the explanation of economic phenomena by using the speech act theory. In addition, it also gives a theoretical reference for small loan companies to understand the borrowers' repayment behavior and improve the debt repayment rate.

6.3 | Practical implications

For companies, first, our results show that a collector with a louder voice and fluent speech can make the borrowers repay loan earlier, which provides a reference for loan companies to recruit and train new employees. For example, loan companies can pay special attention to these voice features when selecting collectors to improve the repayment rate of debt collection calls. Second, the development of artificial intelligence makes it possible to use intelligent voice customer services to reduce labor costs. Thus, our results can also be applied to intelligent voice customer services, providing reference for enterprises to debug the voice metrics of intelligent customer services. Third, our results also show that fast speech speed and higher pitch of collectors cause longer delinquent time. Thus, enterprises can set up independent working space for collectors or set up isolation baffles between collectors to minimize background sound.

Besides, it should also be emphasized in the training that the collectors cannot speak too fast to avoid situations where the borrowers cannot hear clearly and become anxious.

For collectors, on the one hand, they should pay attention to controlling voice metrics when they face different borrowers. For borrowers with less repayment amount, increasing the entropy of energy can effectively increase the loan recovery rate. In addition, controlling the pitch and speed of speech is important for the borrowers with more repayment amount. Appropriate control of loudness is also applicable to increasing the repayment rate of borrowers with a moderate amount of repayment. On the other hand, as for the borrowers, the results show that the longer they say, the longer the delinquent time will be. Therefore, when calling for collection, the collector should explain the intention of the call concisely and express the delinquent time, overdue amount, consequences of nonrepayment, and other information of the borrowers clearly and accurately. Unnecessary words are not allowed to talk with the borrowers on the phone and try to avoid causing dissatisfaction of the borrowers. This can not only save the time of the collector but also shorten the repayment time and increase the repayment rate.

6.4 | Limitation and future research

Our research has several limitations which can be considered in future studies. First, our research is limited to the telephone collection scenario; whether it is applicable to the general field of telephone remains to be verified. Thus, future studies can be carried out in other context of telephone. Second, the data we used are second-hand data rather than field experiments data; hence, other possible alternative interpretations cannot be excluded. In future studies, a more comprehensive experimental method can be considered to eliminate the disadvantage that the second-hand data cannot strongly verify the causality.

ACKNOWLEDGMENTS

The authors acknowledges the support from the National Natural Science Foundation of China (Grant: 71925003) and the support by Funds for Sichuan University to Building a World-class University (Grant: SKSYL2019-01).

DATA AVAILABILITY STATEMENT

The raw data required to reproduce these findings cannot be shared at this time as the data also form part of an ongoing study.

ENDNOTE

- ¹ A loan app named “willful use.” It is a simple, convenient, and fast mobile online lending platform. The borrowers input their personal information and the amount to be borrowed, and the company will issue personal loans after evaluation.

REFERENCES

Apple, W., Streeter, L. A., & Krauss, R. M. (1979). Effects of pitch and speech rate on personal attributions. *Journal of Personality and Social*

- Psychology*, 37(5), 715–727. <https://doi.org/10.1037/0022-3514.37.5.715>
- Austin, J. L. (1962). *How to do things with words*. Oxford University Press.
- Bachu, R. G., Kopparthi, S., Adapa, B., & Barkana, B. D. (2008, June). Separation of voiced and unvoiced using zero crossing rate and energy of the speech signal. In *American Society for Engineering Education (ASEE) Zone Conference Proceedings* (pp. 1–7). American Society for Engineering Education.
- Banks, M., De Silva, A. S. H. T. O. N., & Russell, R. (2015). *Trends in the Australian Small Loan Market*. Melbourne, Australia: Australian Centre for Financial Studies, School of Economics, Finance and Marketing, RMIT University.
- Bernerth, J. B., & Aguinis, H. (2016). A critical review and best-practice recommendations for control variable usage. *Personnel Psychology*, 69(1), 229–283. <https://doi.org/10.1111/peps.12103>
- Black, J. W. (1961). Relationships among fundamental frequency, vocal sound pressure, and rate of speaking. *Language and Speech*, 4, 196–199. <https://doi.org/10.1177/002383096100400402>
- Bressler, S., Masud, S., Bharadwaj, H., & Shinn-Cunningham, B. (2014). Bottom-up influences of voice continuity in focusing selective auditory attention. *Psychological Research*, 78(3), 349–360. <https://doi.org/10.1007/s00426-014-0555-7>
- Brown, B. L., Strong, W. J., & Rencher, A. C. (1974). Fifty-four voices from two: The effects of simultaneous manipulations of rate, mean fundamental frequency, and variance of fundamental frequency on ratings of personality from speech. *Journal of the Acoustical Society of America*, 55, 313–318. <https://doi.org/10.1121/1.1914504>
- Brusa, F., Luo, X., & Fang, Z. (2019). The power of non-monetary incentive: Experimental evidence from p2p lending in China. Available at SSRN. 3405902. <https://www.stern.nyu.edu/sites/default/files/assets/documents/Brusa%20Luo%20Fang-The%20Power%20of%20Non-Monetary%20Incentive.pdf>
- Bursztyn, L., Fiorin, S., Gottlieb, D., & Kanz, M. (2019). Moral incentives in credit card debt repayment: Evidence from a field experiment. *Journal of Political Economy*, 127(4), 1641–1683. <https://doi.org/10.1086/701605>
- Cadena, X., & Schoar, A. (2011). Remembering to pay? Reminders vs. financial incentives for loan payments (No. w17020). National Bureau of Economic Research
- Chebat, J. C., Hedhli, K. E., Gélinas-Chebat, C., & Boivin, R. (2007). Voice and persuasion in a banking telemarketing context. *Perceptual and Motor Skills*, 104(2), 419–437. <https://doi.org/10.2466/pms.104.2.419-437>
- Congdon, W. J., & Shankar, M. (2015). The white house social & behavioral sciences team: Lessons learned from year one. *Behavioral Science & Policy*, 1(2), 77–86. <https://doi.org/10.1353/bsp.2015.0010>
- Coppett, J. I., & Voorhees, R. D. (1985). Telemarketing: Supplement to field sales. *Industrial Marketing Management*, 14(3), 213–216. [https://doi.org/10.1016/0019-8501\(85\)90040-9](https://doi.org/10.1016/0019-8501(85)90040-9)
- Cox, D., & Oakes, D. (1984). In *Analysis of survival data*. London: Chapman and Hall.
- Du, N., Li, L., Lu, T., & Lu, X. (2020). Prosocial compliance in P2P lending: A natural field experiment. *Management Science*, 66(1), 315–333. <https://doi.org/10.1287/mnsc.2018.3216>
- Eijkel, R. V., Niel, H., & Linsink, R. (2011). Group lending and the role of the group leader. *Small Business Economic Journal*, 80(2), 502–522.
- El Ayadi, M., Kamel, M. S., & Karray, F. (2011). Survey on speech emotion recognition: Features, classification schemes, and databases. *Pattern Recognition*, 44(3), 572–587. <https://doi.org/10.1016/j.patcog.2010.09.020>
- Fellner, G., Rupert, S., & Christian, T. (2013). Testing enforcement strategies in the field: Threat, moral appeal and social information. *Journal of the European Economic Association*, 11, 634–660. <https://doi.org/10.1111/jeea.12013>
- Gélinas-Chebat, C., Chebat, J. C., & Boivin, R. (1999, August). Impact of male and female voice cues on consumers' attitudes in telemarketing,

- conference proceedings of 14. In *International Conference of Phonetic Science* (pp. 1577–1580).
- Gil, L., Iddo, G., & Dana, Y. (2014). Spending more time with the customer: Service-providers' behavioral discretion and call-center operations. *Service Business*, 9(3), 427–443.
- Guyer, J. J., Fabrigar, L. R., & Vaughan-Johnston, T. I. (2019). Speech rate, intonation, and pitch: Investigating the bias and cue effects of vocal confidence on persuasion. *Personality and Social Psychology Bulletin*, 45(3), 389–405. <https://doi.org/10.1177/0146167218787805>
- Hermes, N., Lensink, R., & Mehrteab, H. T. (2005). Peer monitoring, social ties and moral hazard in group lending programs: Evidence from Eritrea. *World Development*, 33(1), 149–169. <https://doi.org/10.1016/j.worlddev.2004.09.001>
- Hunjun, Z. A. L. X. Z. (2003). A Cox risk model of double line [J]. *Mathematical Theory and Application*, 1, 107–112.
- Jin, Z., Lin, D. Y., Wei, L. J., & Ying Z. (2003). Rank-based inference for the accelerated failure time model. *Biometrika*, 90, 341–353. <https://doi.org/10.1093/biomet/90.2.341>
- Johnson, C. L., Bartholomae, S., Serido, J., Katras, M. J., & Tobe, E. (2020). Launching a text message intervention to assist student loan borrowers with repayment decisions. *Family and Consumer Sciences Research Journal*, 48(3), 230–244. <https://doi.org/10.1111/fcsr.12349>
- Kagan, J., & Brock, T. (2020). What is consumer credit? Retrieved from <https://www.investopedia.com/terms/c/consumercredit.asp>
- Kalbfleisch, J. D., & Prentice, R. L. (2002). *The Statistical Analysis of Failure Time Data*. New York: Wiley. <https://doi.org/10.1002/9781118032985>
- Karlan, D., Morten, M., & Zinman, J. (2012). A personal touch: Text messaging for loan repayment (No. w17952). National Bureau of Economic Research.
- Ketrow, S. M. (1990). Attributes of a telemarketer's voice and persuasiveness. A review and synthesis of the literature. *Journal of Direct Marketing*, 4(3), 7–21. <https://doi.org/10.1002/dir.4000040304>
- Key, K. F., & Carr Payne, M. Jr. (1981). Effects of noise frequency on performance and annoyance for women and men. *Perceptual and Motor Skills*, Apr(v52), 435–441.
- Klofstad, C. A., Anderson, R. C., & Nowicki, S. (2015). Perceptions of competence, strength, and age influence voters to select leaders with lower-pitched voices. *PLoS ONE*, 10(8), e0133779. <https://doi.org/10.1371/journal.pone.0133779>
- Koomen, A. (1995). Proper telephone usage for increased sales. *American Salesman*, 40(1), 7–11.
- Leech, G. N. (1983). *Principles of Pragmatics*. Longman.
- Ma, Z. S., & Bechinski, E. J. (2009). Accelerated failure time (AFT) modeling for the development and survival of Russian wheat aphid. *Diuraphis noxia* (Mordvilko). *Population Ecology*, 51, 543–548. <https://doi.org/10.1007/s10144-009-0160-6>
- Moulton, S., Collins, J. M., Loibl, C., & Samek, A. (2015). Effects of monitoring on mortgage delinquency: Evidence from a randomized field study. *Journal of Policy Analysis and Management*, 34(1), 184–207. <https://doi.org/10.1002/pam.21809>
- NBC News. (2018) Retrieved from <https://brandongaille.com/26-microfinance-industry-statistics-trends-analysis/>
- Papakpistas, M., Siantikos, G., Giannakopoulos, T., Spyrou, E., & Sgouropoulos, D. (2017). Recognizing emotional states using speech information. In *GeNeDis*, 2016, 155–164.
- Pekonen, P. Are text message reminders effective in debt collection? Randomized controlled trial in debt collection in Finland. <https://aaltodoc.aalto.fi/handle/123456789/14613>
- Ross, J. (1970). On declarative sentences. In R. Jacobs, P. Rossenbaum (Eds.), *Readings in English transformational grammar* (pp. 222–272). Waltham, Massachusetts: Blaisdell.
- Sadock, J. M. (1974). *Towards linguistic theory of speech acts*. New York: Academic Press.
- Searle, J. R. (1969). *Speech acts*. Cambridge University Press.
- Social and Behavioral Sciences Team. 2015 Annual report. Washington, DC: US Office of Science and Technology Policy. Retrieved May 31, 2019. From <https://sbst.gov/assets/files/2015-annualreport.pdf>
- Taylor, A. B., MacKinnon, D. P., & Tein, J. Y. (2008). Tests of the threepath mediated effect. *Organizational Research Methods*, 11, 241–269. <https://doi.org/10.1177/1094428107300344>
- The People's Bank of China. (2020) Retrieved from <http://www.pbc.gov.cn/goutongjiaoliu/113456/113469/4061877/index.html>
- Tian, H., Song, Q. H., & Zhou, X. J. (2014). The simulation of survival time based on accelerated failure time model was applied to detect survival trait loci. *Mathematics in Practice and Theory*, 44(20), 165–171.
- Tigue, C. C., Borak, D. J., O'Connor, J. J. M., Schandl, C., & Feinberg, D. R. (2012). Voice pitch influences voting behavior. *Evolution and Human Behavior*, 33, 210–216. <https://doi.org/10.1016/j.evolhumbehav.2011.09.004>
- Tlidi, F. (2020). Pragmatic analysis of persuasion in Moroccan advertisements (doctoral dissertation, Abdelmalek Essaadi University).
- Visaria, S. (2009). Legal reform and loan repayment: The microeconomic impact of debt recovery tribunals in India. *American Economic Journal: Applied Economics*, 1, 59–81.
- Wharton, T. (2004). Lexical acquisition and pragmatics. *UCL Working Papers in Linguistics*, 16, 323–341.
- Yu, C. Q., Sun, X. L., & Wang, Z. Z. (2005). A study on the attitude of over-paid credit card holders to the telephone collection mechanism of the issuing bank. *Marketing Comments*, 2(4), 395–422.

How to cite this article: Huang, Z., Yang, Y., Liao, C., & Du, P. (2021). How to say? Voice analytics of debt collection strategies. *Managerial and Decision Economics*, 1–14. <https://doi.org/10.1002/mde.3441>

APPENDIX A: DETAILS OF BORROWERS' BACKGROUND

Before officially obtaining the loan, potential borrowers need to provide the loan store with a copy of ID card, basic demographic information, and a third-party contact person. Subsequently, the loan store will confirm the loan and demographic information with potential borrowers through calls to assess the credit risk and complete the risk pricing. The final loan quota shall not exceed the risk pricing quota, and the loan amount will be adjusted according to the credit risk level of the borrowers. In general, the loan amount of borrowers with high credit risk will be reduced.

The potential borrowers will be informed of the consequences of delinquent before signing the loan contract. For example, in the



FIGURE A1 The details of the app [Colour figure can be viewed at wileyonlinelibrary.com]

predue period, interest, service fees, or late fees are usually required to be charged. If the delinquent account has not been repaid for a long time, the loan store will contact a third-party guarantor to assume the debt or require the borrowers to assume other responsibilities. The details of the app are shown in Figure A1.

APPENDIX B: DETAILS OF VOICE MINING

Audio data storage. The audio data including the voice of collectors and borrowers are stored in left and right channels, respectively, which facilitate for audio clipping. The audio signal is an analog signal. Its sampling frequency is set as 44,100 Hz, and the quantization bit is 16 bits. Subsequently, we have performed noise reduction processing on audio files. Finally, the audio files are stored into two wav-type categories according to the voice of collectors and borrowers.

Automatic speech recognition (ASR). We use the API of Tencent-ASR system to convert audio files into text files with time stamps. We use ASR, which provides very accurate audio recognition, to indicate the conversation content labeled with time of collectors and borrowers. The operation principles of ASR are shown in Figure B1.

Voice feature extraction. We use *pyAudioAnalysis* tool to extract voice metrics whose library is written in Python. According to *pyAudioAnalysis*, we first divide the audio signals into short-term windows (frames), then calculate the 34 voice metrics for each frame with two overlapping times and finally extract the means and standard deviations of 34 metrics for each sentence. The voice metrics can be divided into three categories. The first one is time domain features, such as *zero crossing rate* (the rate of sign-changes of the signal during the duration of a particular frame), *energy* (the sum of squares of the signal values, normalized by the respective frame length), and *entropy of energy* (the entropy of subframes' normalized energies. It can be interpreted as a measure of abrupt changes). The second one is frequency-domain features which are based on the magnitude of discrete Fourier transform (DFT). This category includes *spectral centroid* (the center of gravity of the spectrum), *spectral spread* (the second central moment of the spectrum), *spectral entropy* (entropy of the normalized spectral energies for a set of subframes), *spectral flux* (the squared difference between the normalized magnitudes of the spectra of the two successive frames), *spectral rolloff* (the frequency below

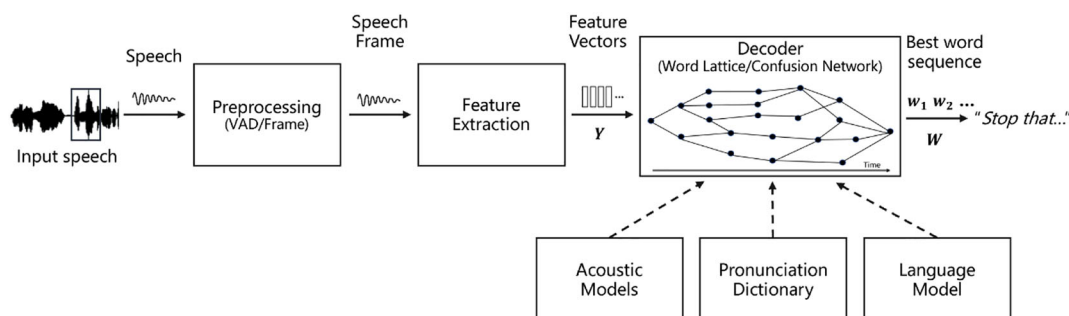


FIGURE B1 The operation principles of automatic speech recognition system (ASR) [Colour figure can be viewed at wileyonlinelibrary.com]

which 90% of the magnitude distribution of the spectrum is concentrated), and 12 *chroma vectors* (a 12-element representation of spectral energy in which the bins represent the 12 equal-tempered pitch classes of Western-type music) and chroma deviation (the standard deviation of the 12 chroma coefficients). The last category is the 13 Mel-frequency cepstral coefficients (MFCCs); it belongs the cepstral domain results after applying the inverse DFT on the logarithmic spectrum and forms a cepstral representation where the frequency bands are not linear but distributed according to the mel-

scale. The independent variable that is entropy of energy and the mediator that is energy are produced by this way.

Pitch and loudness are extracted by the tool of *librosa* in Python. Just like the *pyAudioAnalysis* tool, we separate the audio signal into frames then calculate these two variables for each frame with two overlapping times and extract the means and standard deviations of pitch and loudness. Speed is calculated according to the number of words per second.